Stress Testing

User Guide — CDS-in-Stress, Flood Classifiers & Portfolio Scenarios

MKM Research Labs

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Contents

1 Introduction

Stress testing answers the question: *what would happen to our PRS positions if a specific storm hit the catchment?*

The platform uses its catalogue of 20,000 synthetic storm sequences to run hourly scenarios. For each hour of a 168-hour storm, the system computes the gauge water level, the probability of flooding, and the resulting P&L impact on every affected trade. This produces a time-resolved view of how the portfolio behaves under stress.

1.1 CDS-in-Stress Concept

In normal mark-to-market, PRS positions are valued using CDS-style analytics (hazard rate → fair spread → MTM). Under stress, the position transitions from a credit instrument to a **cash-settled physical payout**:

$$\text{cash price} = N \times P(\text{flood}) \times d$$

where N is the notional, $P(\text{flood})$ is the real-time flood probability, and d is the direction ($+1$ payer, -1 receiver). The stress P&L is the difference between this cash price and the normal-market MTM.

2 Gauge-Level Stress

2.1 Workflow

1. **Select a gauge** from the dropdown. Only gauges with open trades are shown.
2. **Select a storm** from the catalogue. The list shows storms that breached alert level at this gauge, sorted by severity (catastrophic first).
3. **Run the scenario.** The server computes the hourly P&L trajectory over 168 hours.

2.2 What the Charts Show

The stress tab displays three sub-charts:

Flood Probability A line chart of $P(\text{flood})$ vs. hour. Shows how the probability rises as water level approaches the trigger threshold, peaks, and then recedes.

Stress P&L A line chart of portfolio stress P&L vs. hour. Shows the accumulated cash-price impact of the storm.

Probability Surface A 3D surface of $P(\text{flood})$ vs. water level vs. hour. Useful for understanding how sensitivity changes at different water levels.

3 The Hourly Scenario

For each hour h in the 168-hour window:

1. **Water level** w_h is read from the storm's gauge response hydrograph (pre-computed from the storm catalogue).
2. **Flood probability** $p_h = f(w_h, h)$ is computed using either a trained GBM classifier or the analytical flood polynomial.
3. **Cash price** per trade: $C_h = N \times p_h \times d$.
4. **Stress P&L** per trade: $\Pi_h = C_h - \text{MTM}$.
5. **Portfolio stress P&L**: $\sum \Pi_h$ across all affected trades.

3.1 Knock-Out Behaviour

When the water level exceeds a trade's trigger threshold:

- The trade "knocks out" — the protection buyer receives full notional.
- $P(\text{flood}) = 1.0$ from the knock-out hour onward.
- Stress P&L jumps to $N - \text{MTM}$ (full notional minus the original mark).
- The trade is no longer re-priced in subsequent hours.

In the chart, the P&L line flattens after knock-out because no further P&L change can occur.

3.2 Hydrograph Source

The hourly water levels come from the gauge response stored in `gaugets/{gauge_id}.json`. The storm's `peak_level_m` and duration are used to scale the hydrograph template. If the gauge time series is not available, the system falls back to a flat hydrograph (constant water level for 168 hours).

4 Flood Probability Models

Two models can compute $P(\text{flood})$. The classifier is preferred when available because it captures non-linear effects.

4.1 GBM Classifier (Preferred)

A **Gradient Boosting Machine** trained per gauge using historical storm simulations. The model takes four features:

Feature	Description
`log_h_s`	Log of water level relative to severe threshold: $\ln(w/w_{\text{severe}} + \varepsilon)$
`log_t_end`	Log of time remaining in the 168-hour window: $\ln(168 - h + \varepsilon)$
`delta_log_h`	First derivative of log water level (rate of rise/-fall)
`delta2_log_h`	Second derivative (acceleration of rise/fall)

The classifier outputs a probability in $[0, 1]$. It is more accurate than the polynomial model because it learns the non-linear relationship between water level dynamics and flood outcomes from the Monte Carlo data.

4.2 Analytical Flood Polynomial (Fallback)

When no classifier is trained for a gauge, the system uses a polynomial approximation:

$$P(\text{flood}) = \sigma\left(a_0 + a_1\frac{w}{w_s} + a_2\left(\frac{w}{w_s}\right)^2 + a_3\frac{h}{168}\right)$$

where σ is the logistic sigmoid and the coefficients are calibrated to the Thames. This model is smooth but cannot capture the non-linear dynamics learned by the classifier.

4.3 When to Use Which

	GBM Classifier	**Flood Polynomial**
Accuracy	Higher (trained on data)	Lower (generic fit)
Availability	Requires training	Always available
Speed	Fast (tree lookup)	Instant (formula)
Per-gauge	Yes (gauge-specific)	No (one model for all)

5 Portfolio-Level Stress

The Portfolio Stress tab runs a single storm across **all gauges** simultaneously, computing the aggregate impact.

5.1 Workflow

1. The tab lists all storms in the catalogue, sorted by the number of gauges that breached severe (most impactful first).
2. Select a storm.
3. The server runs the scenario for every gauge that has open trades and returns per-gauge results.

5.2 Output

For each gauge, the response includes:

- Peak water level and threshold classification (alert, warning, severe, or clean).
- $P(\text{flood})$ at peak.
- Stress P&L contribution.
- Number of affected trades.
- Full 168-hour hydrograph.

The portfolio summary shows:

- Total stress P&L across all gauges.
- Number of gauges at each severity level.
- Worst-hit gauge and its contribution.

5.3 Drill-Down

Clicking a gauge in the portfolio stress results switches to the Stress tab (single gauge) and pre-selects that gauge and storm, allowing detailed hourly analysis.

6 Probability Surface

The probability surface is a 3D visualisation of $P(\text{flood})$ as a function of water level and hour. It is generated by evaluating the classifier (or polynomial) across a grid of water levels and time points.

6.1 How to Read It

- The x-axis is time (0–168 hours).
- The y-axis is water level (metres).
- The z-axis (colour) is $P(\text{flood})$.
- The actual storm trajectory is a path across this surface.

The surface reveals the classifier's learned behaviour:

- At low water levels, $P(\text{flood}) \approx 0$ regardless of hour.
- At high water levels, $P(\text{flood}) \approx 1$ regardless of hour.
- The transition zone (e.g. 0.2–0.8 probability) is where the storm's trajectory determines the outcome.
- Late in the window (hour 140+), the probability decreases because there is insufficient time for a full flood event to develop.

7 Classifier Training

7.1 Training a Classifier

To train a GBM classifier for a gauge:

1. Navigate to the Classifiers tab.
2. Select the gauge.

3. Click **Train**. The system builds a feature matrix from all storm scenarios at this gauge and trains a GBM model.

4. Training typically takes 10–60 seconds per gauge.

5. The trained model is saved as a `.joblib` file in `classifiers/{gauge_id}.joblib`.

7.2 Batch Training

The **Train All** button trains classifiers for all gauges with sufficient historical data. This is an asynchronous job; progress is shown in the UI.

7.3 AUC Metric

After training, the classifier's quality is measured by the **Area Under the ROC Curve** (AUC):

> 0.95 Excellent — very reliable flood probability estimates.

0.85–0.95 Good — reliable for stress testing.

< 0.85 Fair — consider retraining with more data or using the polynomial fallback.

7.4 When to Retrain

Classifiers should be retrained when:

- The storm catalogue is regenerated (new storm control parameters).
- Gauge hazard data is updated.
- The AUC drops below 0.85 (indicates the model is stale).

8 Parameter Reference

Parameter	Default	Increase	Decrease
`STRESS_STORMS_MIN_COUNT`	50	Excludes gauges with few storms	Includes more gauges
`STORM_DEFAULT_DURATION`	168 h	Longer fallback hydrograph	Shorter fallback
`DEFAULT_PEAK_POSITION`	0.5	Late peak (delayed climax)	Early peak (front-loaded)
`LOG_END`	168.0	Classifier feature scaling	Narrower time window
`LOG_EPS`	10^{-8}	Less numerical sensitivity	More sensitivity at zero

Parameter	Default	Increase	Decrease
BUMP_1BP	0.0001	Larger sensitivity step	Finer stress resolution

Document History

Version	Date	Author	Description
1.0	10-Apr-2026	MKM Research Labs	Initial release